

Chapter 3: Plasticity of functional traits and its evolutionary constraints in *Nemophila menziesii*

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Author Contributions: S.J.M. conceived and obtained funding for this work. D.G., L.K., H.E.P., and S.J.M. designed the experiment, supervised and conducted the breeding of source seeds, established field sites, trained participating UCSB undergraduate students, collected greenhouse and field data, and interpreted the analyses. H.E.P. performed data analysis and interpretation with the help of S.J.M. H.E.P. led the preparation of the manuscript, with S.J.M. revising and editing the text. All authors reviewed and approved the final version of the manuscript.

Conflict of Interest: The authors declare no conflicts of interest.

Data Availability: All data supporting the findings of this study are openly available in the GitHub repository:

Funding:

This project was supported by National Science Foundation DEB-1655727 to SJM. The Schuyler Greenhouse, where seeds were reared, was funded by Dr. A.H. Schuyler and NSF Grant OIA-0963547. The LED grow light used was supported by the UCSB Green Initiative Fund and NSF Grant OIA-0963547.

Acknowledgements:

We thank Amber Nashoba for assisting with the initial experimental design (cf. to previous work by Ruth G. Shaw) and for collecting seeds from field populations. We are also grateful to Ivana Gomez, who participated in field work and in the breeding of the parental generation, and our 2023 field technicians, Lynn Jung, Riki Radliff, and Ally Wetzel. We also thank Cameron Hannah-Bick for her invaluable assistance with rearing *N. menziesii* seeds in the greenhouse. We greatly appreciate the many undergraduate researchers who contributed to greenhouse work, sample processing, and/or data management. We are particularly thankful to NSF for supporting the participation of the following students by providing Research Experiences for Undergraduate supplements: Michelle Colvin, Bergen Foshay, Cayenne Gularte, Lynn Jung, Jonathan Lam, Grayson Prater, and Rishima Tewari, and for supporting Karen Presburger and Greg Schiller on a Research Experiences for Teachers award. We also extend our sincere gratitude to Ruth Shaw for her invaluable guidance and support throughout the course of this project.

Keywords: Phenotypic plasticity, estimated breeding values, fitness, corolla diameter, water use efficiency, specific-leaf area, mean seed mass, *Nemophila menziesii*, selection

Abstract:

Phenotypic plasticity allows genotypes to adjust trait expression in response to environmental variation, but such changes do not necessarily increase fitness. We examined the magnitude, direction, and fitness consequences of plasticity in four functional traits (specific leaf area, $\delta^{13}\text{C}$, mean seed mass, and corolla diameter) in two environmentally distinct years in four pedigreed populations (Angelo Coast, Bodega Bay, Blue Oak, and Hastings Reserve) of the California annual wildflower *Nemophila menziesii*. By evaluating the phenotype and performance of siblings in both years, we estimated within-genotype plastic responses and tested whether plasticity was associated with fitness and exhibited genetic variation. We detected substantial trait-specific plasticity, with some traits exhibiting greater sensitivity to environmental variation than others. However, there was little evidence of genotype-by-year interactions, indicating that plastic responses were similar among paternal families. Although paternal families differed in how trait values changed between years, these differences in plasticity were rarely associated with consistent fitness advantages. Associations between plasticity and fitness were weak, inconsistent, and dependent on the trait, population, and environmental context. Moreover, plasticity exhibited low genetic variation, suggesting limited evolutionary potential of plasticity. Our results indicate that while all of our focal traits exhibited significant plasticity at the population level, genetic variation in plasticity within populations was not high, and plasticity did not reliably enhance fitness. More broadly, this work highlights the importance of integrating plasticity, fitness, and genetic variation to predict how populations will respond to environmental change.

Introduction:

Phenotypic plasticity, or environmentally induced changes in trait expression, enables plants to respond to variable environmental conditions (Schlichting, 1986; Pigliucci, 2001;

West-Eberhard, 2003). Plastic responses may be adaptive when they allow individuals to adjust to environmental conditions within and between generations (West-Eberhard, 1989). In contrast, robustness (homeostasis, or canalization) refers to the relative stability of trait expression across environmental conditions, whereby phenotypes remain consistent despite environmental variation (Kaneko, 2012). When genetic variation exists in trait expression or in genotype-specific responses to environmental variation, selection can act on this variation, shaping evolutionary trajectories (West-Eberhard, 1989).

Although phenotypic plasticity can buffer populations against environmental change, these responses do not necessarily result in an increase in fitness in all environments and may impose costs that constrain evolutionary responses (DeWitt et al., 1998; Pigliucci, 2005; Ghalambor et al., 2007; Reed et al., 2011; Murren et al., 2015; Yi and Dean, 2016). Thus, determining whether plasticity is adaptive in a particular population or taxon requires evaluating not only the phenotypic response of genotypes to different environmental conditions, but also correlating plasticity with genotypic fitness within and across these conditions. Despite being extensively studied, it remains unclear whether plastic responses of functional traits affect fitness in natural populations and whether there is sufficient genetic variance in plasticity itself for plasticity to evolve (Walter et al., 2023; de la Mata et al., 2024). One reason this question remains unresolved is that plastic responses are often trait-specific, as different traits are mediated by distinct developmental and physiological pathways (Bradshaw, 1965; Scheiner, 1993; Sultan, 2000; Fusco and Minelli, 2010; Klingenberg, 2019; Anderson et al., 2021). Moreover, determining whether plasticity has the potential to evolve requires evaluating whether genotypes differ in their responses to environmental variation (Via and Lande, 1985; Eriksson et al., 2023). Because genotype-by-environment interactions reflect differences among paternal

families in the shape or slopes of their reaction norms, they indicate genetic variation in plasticity and may influence the evolution of trait means across heterogeneous environments (Falconer, 1952; Via and Lande, 1985; Scheiner and Lyman, 1989; Scheiner, 1993; Nussey et al., 2007).

Temporal environmental heterogeneity can result in selection on plastic responses when genotypes differ in their responses across environments, particularly in populations experiencing strong interannual variation (Via and Lande, 1985). In Mediterranean-climate systems such as coastal California, precipitation and temperature often vary substantially among years, creating marked interannual differences in growing conditions (Dettinger et al., 2011; Polade et al., 2017). *Nemophila menziesii* H. and A. (Hydrophyllaceae), an annual wildflower native to California, is well suited for examining plastic responses to such interannual variation and the fitness consequences of these responses because its life history and reproductive success are strongly influenced by environmental conditions during development (Cruden, 1972; Ganders, 1978; Platenkamp and Shaw, 1993; Andersson, 1994; Shaw et al., 1998; Barr, 2004; Gomez and Shaw, 2006; McCall, 2008; McCall and Barr, 2012).

In this study, we raised four pedigreed populations of *N. menziesii* at their home site in two different years, measuring morphological traits and physiological traits, as well as fecundity. Based on the set of functional traits identified by Nicotra et al. (2010) and building on previous research in *N. menziesii* (Cruden, 1972; Ganders, 1978; Platenkamp and Shaw, 1993; Andersson, 1994; Shaw et al., 1998; Barr, 2004; Gomez and Shaw, 2006; McCall, 2008; McCall and Barr, 2012), we selected four focal traits for investigation: specific leaf area (SLA), delta carbon 13 ($\delta^{13}\text{C}$), mean seed mass, and corolla diameter. These traits represent key aspects of plant function, including resource acquisition, water-use efficiency, and reproductive investment.

By growing siblings in two environmentally distinct years that differed substantially in precipitation, we estimated trait-specific and multivariate plasticity and tested whether families differed in their phenotypic responses between years. Additionally, we tested for genetic variation in plasticity by evaluating Genotype $\times$ Year (hereafter, G $\times$ Y) interactions, which occur when paternal families differ in how their phenotypes respond to environmental variation. We also evaluated whether trait-specific or multivariate plasticity was associated with fitness within each population. Conceptual predictions for these relationships are illustrated in Figure 1, which shows how different reaction norm patterns may relate to fitness between years. Finally, by integrating these analyses with estimates of trait-specific additive genetic variance and covariance reported in Chapter 2 (Payne et al., *in prep*), we evaluated how plastic responses and underlying genetic variation may influence the evolvability of traits within each population.

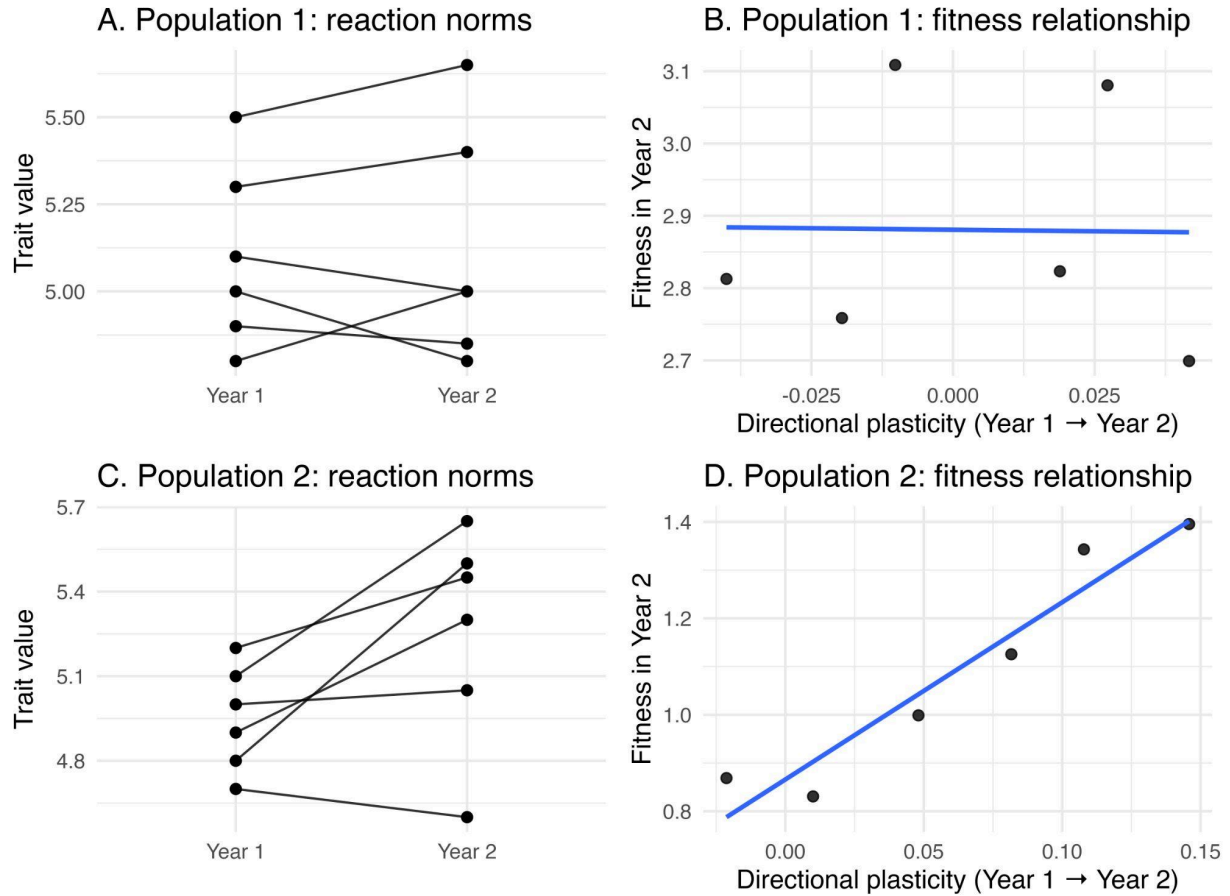


Figure 1. Examples of trait-specific reaction norms and their relationship to fitness in each of two populations, each observed in two environmentally distinct years. Panels A and C illustrate paternal family-specific reaction norms for a focal trait observed in two different populations. Panels B and D show the corresponding hypothetical relationships between directional plasticity in that trait and fitness in Year 2; each point represents the directional plasticity (x-axis) and mean fitness (y-axis) of one of the six paternal families observed in Population 1 (Panel B) and in Population 2 (Panel D). Positive values of the x-axis indicate paternal families whose phenotypic values increased between Years 1 and 2; negative values indicate paternal families whose phenotypic values declined between years. In Population 1 (A and B), paternal families vary in trait expression between years, but plasticity is not associated with fitness in Year 2, indicating no evidence of a significant fitness effect of this trait's plasticity under the environmental conditions experienced in Year 2. In Population 2 (C and D), paternal families differ in the magnitude and direction of trait change, and greater “positive” plasticity (i.e., greater increases in phenotype between Years 1 and 2) is associated with higher fitness in Year 2, consistent with the inference that phenotypic increases in the trait's value between Year 1 and Year 2 are adaptive.

Methods:

Study Species: *N. menziesii* is a gynodioecious, outcrossing, annual wildflower distributed along the west coast of North America (Cruden, 1972). Although multiple varieties are recognized based on floral morphology, all individuals within this study were treated collectively as *N. menziesii* without differentiating them by variety.

Experimental design–Study populations: This study was conducted at four UC Natural Reserve System (UCNRS) sites where *N. menziesii* naturally occurs. These sites differ in latitude and, consequently, in long-term climatic conditions, including mean annual and seasonal temperature, precipitation, and vapor pressure deficit (Table 1). All four sites experienced interannual variation in temperature, precipitation, and vapor pressure deficit between the two years (2022 and 2023, hereafter referred to as Years 1 and 2) examined in this study. From north to south, these sites are Angelo Coast Range Reserve (Angelo Coast, AC), Bodega Bay Marine Laboratory (Bodega Bay, BB), Blue Oak Ranch Reserve (Blue Oak, BO), and Hastings Natural History Reservation (Hastings Reserve, HR; Figure 2).

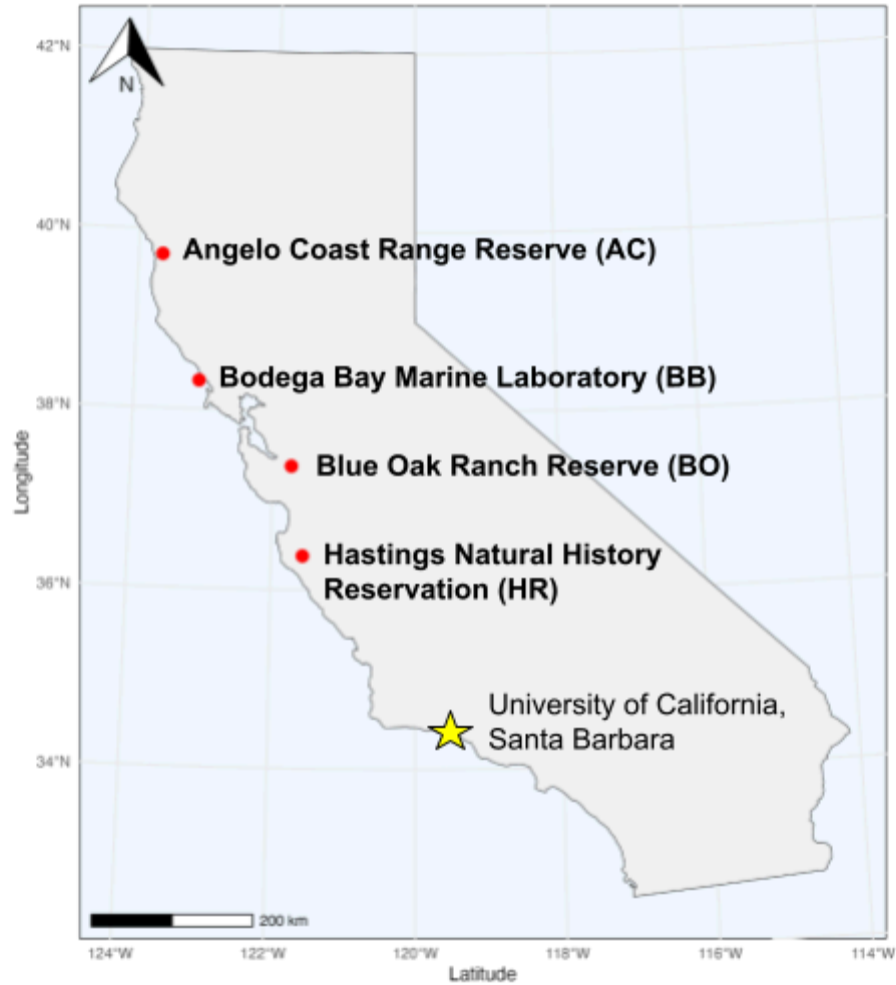


Figure 2. Locations of field sites of the four pedigreed populations observed under field conditions: Angelo Coast Range Reserve (AC), Bodega Bay Marine Lab (BB), Blue Oak Ranch Reserve (BO), and Hastings Natural History Reservation (HR). The Angelo Coast Range Reserve is ~500 kilometers north of the most southern site, Hastings Natural History Reservation.

Table 1. Mean annual and spring temperature, cumulative annual and spring precipitation, latitude, longitude, and elevation for each study site: Angelo Coast (AC), Bodega Bay (BB), Blue Oak (BO), and Hastings Reserve (HR). Climatic data are based on 30-year climate normals (means) ($\pm$ SD) recorded from 1991-2020 (Wang et al., 2016). Mean spring temperature is calculated as the mean of March, April, and May mean daily temperatures. Spring vapor pressure deficit (VPD) is calculated based on mean spring temperature and mean spring relative humidity. VPD was calculated from air temperature and relative humidity using standard equations, where saturation vapor pressure was estimated as a function of temperature and scaled by relative humidity to obtain actual vapor pressure (Sadler and Evans, 1989).

Sit e	Latitu de	Longitu de	Annual PPT(mm)	Spring PPT (mm)	Annual Temp (°C)	Spring Temp (°C)	Spring VPD (kPa)	Elevati on (m)
A C	39.746 9	-123.638 0	2135.6 ± 637.1	541.4 ± 255.8	12.3 ± 0.58	10.5 ± 1.1	0.44	567
B B	38.314 6	-123.069 4	910.5 ± 262.9	222.7 ± 118.6	12.43 ± 0.53	11.7 ± 0.9	0.52	2
B O	37.387 0	-121.715 8	670.4 ± 203.5	185.0 ± 101.7	14.77 ± 0.58	12.4 ± 1.1	0.38	728
H R	36.378 3	-121.566 2	729.9 ± 222.5	187.5 ± 100.9	14.78 ± 0.59	12.4 ± 1.0	0.49	603

Seed sampling: At each site, seeds were collected in spring 2018 from a minimum of 200 maternal plants of *N. menziesii*. All seeds collected from a given maternal plant were stored in a paper coin envelope (one maternal family per envelope) and placed in a plastic zip-lock bag with silica gel to keep the seeds dry. To reduce the likelihood of sampling closely related individuals, seeds were collected from maternal plants growing at least 1-m apart.

Breeding Design: The breeding design used to create pedigreed seeds in each population follows the protocol from Payne et al. (*in prep*). In the greenhouse from August 2018 to July 2019, offspring from each field-sampled maternal plant were reared and hand-pollinated in cohorts using a paternal half-sib breeding design to create a pedigreed population of seeds (see S1 for details of plant cultivation, greenhouse conditions, and the numbers of maternal and paternal families of pedigreed seeds produced for each population). In this half-sibling breeding design, the pollen of each donor (sire) was used to pollinate three recipients (dams), each of which was derived from a seed produced by a distinct field-collected maternal family. In each population, 50 individuals were randomly assigned as pollen donors and 150 as pollen recipients. Because greenhouse space was limited, pollinations were carried out in seven successive cohorts (see

supplementary material for details on cohorts and pollination protocol).

To confirm that insect-mediated pollination was not occurring in the greenhouse, 13 plants (3 or 4 plants per population) with all flowers emasculated during development were interspersed among experimental pollen recipients in Cohort 4. Fruit production on these control plants was monitored, and the near absence of fruit set indicated that unintended pollination was negligible ($N = 1,651$ flowers, 0.02% of which produced fruit; greenhouse conditions can be found in the supplemental material Table S1).

As the fruits of the maternal plants matured, they were harvested and placed in coin envelopes and stored with silica gel in sealed plastic bags. These pedigreed seeds were stored in a dark, cold room, suspending their development until fall 2021, when the field experiment began. These pedigreed seeds were sown in the field at their respective population of origin in two successive years.

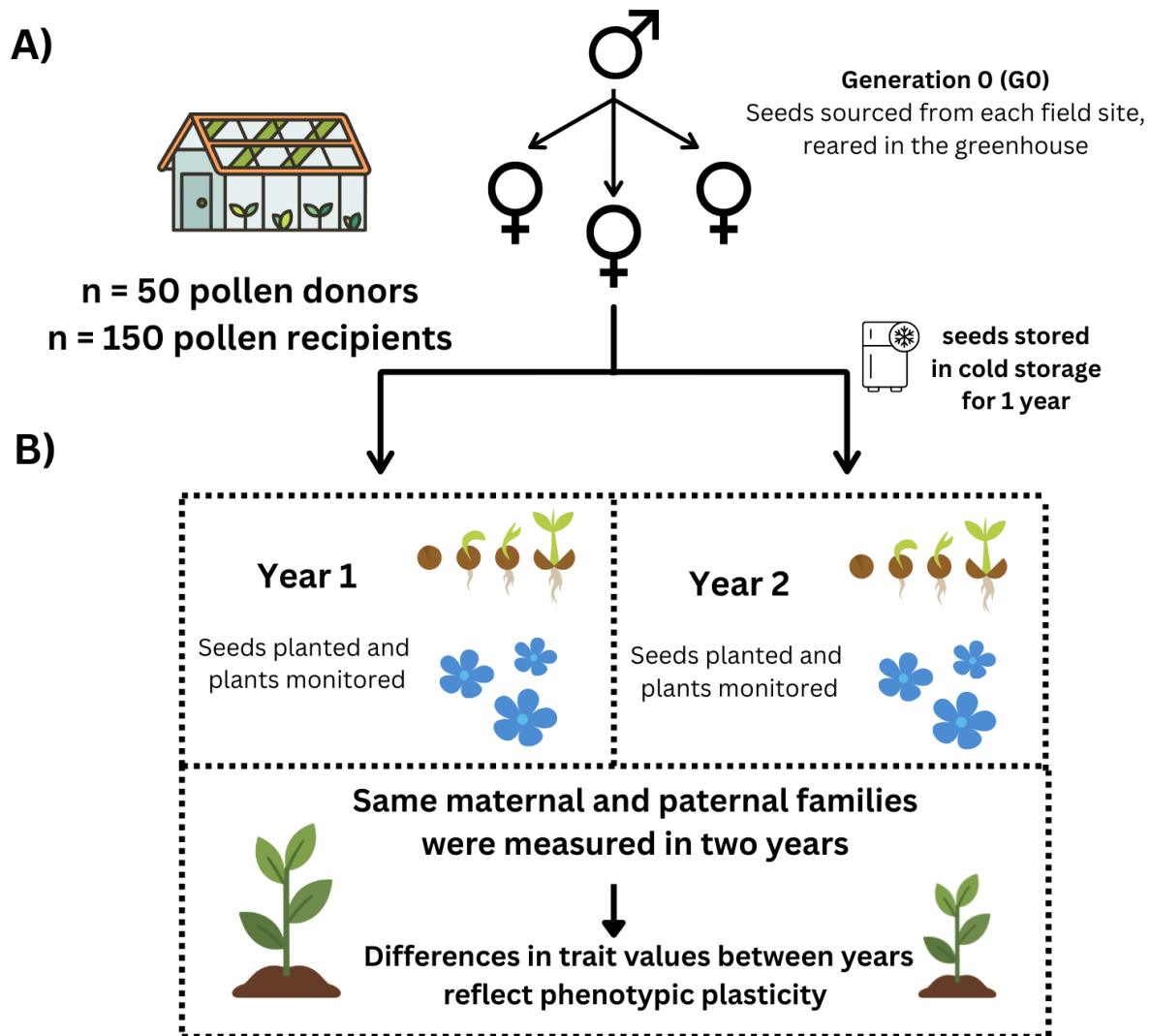


Figure 3. Experimental design for quantifying plasticity and estimating parental effect on fitness between years. A) Seeds of *N. menziesii* were collected from each field site and grown in the greenhouse (Generation 0). Upon flowering, individuals were randomly designated as pollen donors ($n = 50$) or pollen recipients ($n = 150$). Controlled crosses were conducted using a half-sibling breeding design, in which each donor was used to pollinate three recipients, generating a total of ~ 150 pedigreed maternal families per population. B) Pedigreed seeds were divided into two groups, each containing representatives from all maternal families. One group was planted at the population of origin in the fall of Year 1 (2021) and monitored through the following spring, while the remaining seeds were held in cold storage. In the fall of Year 2 (2023), stored seeds from the same maternal families were planted and monitored under field conditions. This design allowed closely related pedigreed families to be evaluated under field conditions in two years, enabling comparisons of trait expression in two different environments to quantify plasticity and assess genotype $\times$ year ($G \times Y$) interactions.

Experimental field populations — Year 1: In Year 1, seeds from each maternal family within each population were divided into two groups: one group was planted at the population of origin in fall 2021, while the remaining seeds were retained in cold storage until fall 2022 (Figure 4). At each site, three blocks were established, and within each block, 3-4 parallel transects were established approximately 5 meters apart. Each transect was divided into segments of 1-m length, spaced 1-m apart. The number and configuration of transects differed among sites due to the area, shape, and slope of suitable substrate: at AC, 9 transects were established, each with 36 1-m segments; at BO, 12 transects were established, each with 23-44 segments, and at HR, 9 transects were established, each with 46 segments.

Seeds from each greenhouse-pollinated maternal plant were randomly assigned to one 1-m segment within each block. Along the length of each segment, 8 seeds (10 at AC) from the assigned maternal family were sown at 10-cm intervals.

At BB, the patchy distribution of suitable habitat for *N. menziesii* precluded the use of transects. Instead, 156 georeferenced planting locations were established adjacent to individuals of *Eriogonum latifolium* and/or *Lupinus arboreus*, which act as nurse plants for *N. menziesii*. At each location, approximately three 1-m segments were placed near a given nurse plant, with each segment randomly assigned to a different maternal family. At BB, each maternal family was sown in three 1-m segments.

In total, 13,194 seeds were sown and individually monitored in Year 1 (3,210 seeds at AC; 3,528 seeds at BB; 3,168 seeds at BO; and 3,288 seeds at HR, see supplementary material for additional planting details). The half-sibling breeding design enabled the measurement of the paternal family-specific plastic responses of morphological and physiological traits, as well as maternal fecundity, to environmental conditions that differed between the two years of our field

experiment. With these measures of genotypic performance, we also examined the relationship between plasticity and the estimated parental effects on fitness of each paternal family in each year (Payne et al. *in prep*).

Focal traits recorded in each 1-m segment: Components of fitness were recorded for each 1-m segment. Survivorship was recorded as the proportion of the sown seeds that survived to flower in a maternal family's 1-m segment. Morphological and physiological traits, and reproductive success, were measured in each 1-m segment by choosing flowering individuals from which to record the following traits: corolla diameter (which is strongly phenotypically correlated with corolla area in all populations; AC: $r = 0.95$, $N=319$, $p = 2.2e-16$; BB: $r = 0.96$, $N = 370$, $p = 2.2e-16$; BO: $r = 0.96$, $N = 339$, $p = 2.2e-16$; HR: $r = 0.95$; $N = 406$, $p = 2.2e-16$), specific leaf area, $\delta^{13}C$ (a proxy for integrated water-use efficiency), lifetime fruit production, mean seeds per fruit, and mean seed mass of the offspring (see Table 3 for descriptions of how each trait was measured and the units of measurement).

Specific leaf area required destructive sampling, and thus was measured on a different flowering individual than traits that did not require destructive sampling (corolla diameter, $\delta^{13}C$, mean seed mass, total fruit production, mean seeds per fruit). The non-destructively sampled plants for which most data were recorded are referred to here as “fitness” plants; the destructively sampled individuals are referred to as “non-fitness” plants. If a selected fitness plant died before survival to flowering, another flowering individual within the same segment was randomly selected as a replacement.

From the fitness plants, all mature fruits were counted, collected, and then taken to the lab to remove and to count the number of mature seeds per fruit. Open (dehiscent) fruits and intact (closed) fruits were stored separately so that the mean number of filled seeds produced per

fruit could be estimated using only the seed counts from intact fruits. Seeds from intact fruits were counted only if they appeared to be “filled” (fully mature). From each fitness plant, up to 30 seeds from filled fruits were weighed from each sampling trip (once every two weeks) to estimate the mean individual seed mass of the seeds representing each field-sown maternal and paternal family.

Table 3. Summary of the traits measured in this study and how field populations were sampled to record phenotypic values. Each trait was measured for either fitness plants (F) or non-fitness plants (NF) in both spring 2022 and spring 2023 for all populations. Specific-leaf area, which required destructive sampling, was recorded for non-fitness plants (one adult plant per 1-m segment; all other traits were measured on the fitness plant). See Methods section (*Focal traits recorded in each 1-m segment*) for additional information.

Plant Traits	Description of Trait Recorded	Plants Measured
Corolla diameter	Photos were taken in the field of the first flower produced by each plant that was observed to flower; images included a ruler for calibration. ImageJ (Schneider et al., 2012) was used to estimate corolla diameter and area in each photo. Units are in millimeters.	F
Specific leaf area	$SLA = \frac{\text{leaf area}}{\text{dry leaf weight}}$ One leaf per non-fitness plant was collected, stored in a coin envelope placed in silica gel in a zip-lock plastic bag, returned to the lab, dried in a drying oven (at 60°C) for 24 hours, and weighed. Standardized photographs were taken in the lab and leaf area was measured using ImageJ. Units of SLA are in square millimeters per milligram.	NF
δ13C	Carbon isotope composition was measured from above-ground stem tissue. Samples were dried, pulverized, and analyzed at the UC Davis Stable Isotope Facility using an elemental analyzer coupled to an isotope ratio mass spectrometer (EA-IRMS) (Brandeberry, 2020). Values were reported in parts per thousand (‰) relative to the Vienna Pee Dee Belemnite (VPDB) standard and are typically negative in plant tissues due to discrimination against ¹³ C during photosynthesis.	F
Mean seed mass	For each fitness plant, up to 30 seeds were weighed at each biweekly sampling interval to estimate the mean individual seed mass produced by each maternal and paternal family. Units are in milligrams.	F
Total fruit	Total lifetime fruit production of an individual plant (including both dehiscent and	F

production	intact fruits). Units are the number of fruits.	
Mean seeds per fruit	The mean number of filled seeds produced per fruit for each fitness plant. The mean was calculated as the sum of all seeds produced by all intact fruits available from a plant, divided by the number of intact fruits. Units are the number of seeds.	F

Experimental field populations — Year 2: In Year 2 (fall 2022), the parental seeds reserved under cold storage were planted at each field site using a similar spatial design as in Year 1 (Figure 3, Figure 4). The transects and one-meter segments established in Year 1 were reused in Year 2, however, to conserve space, each segment contained two maternal families of five seeds each, rather than a single maternal family of eight seeds. The locations of all families were randomized. A total of 12,545 seeds were sown (3,195 seeds at AC; 2,205 seeds at BB; 3,090 seeds at BO; and 4,055 seeds at HR, see supplementary material for additional planting details).

The same traits as described in Table 3 were recorded every two weeks throughout Year 2.

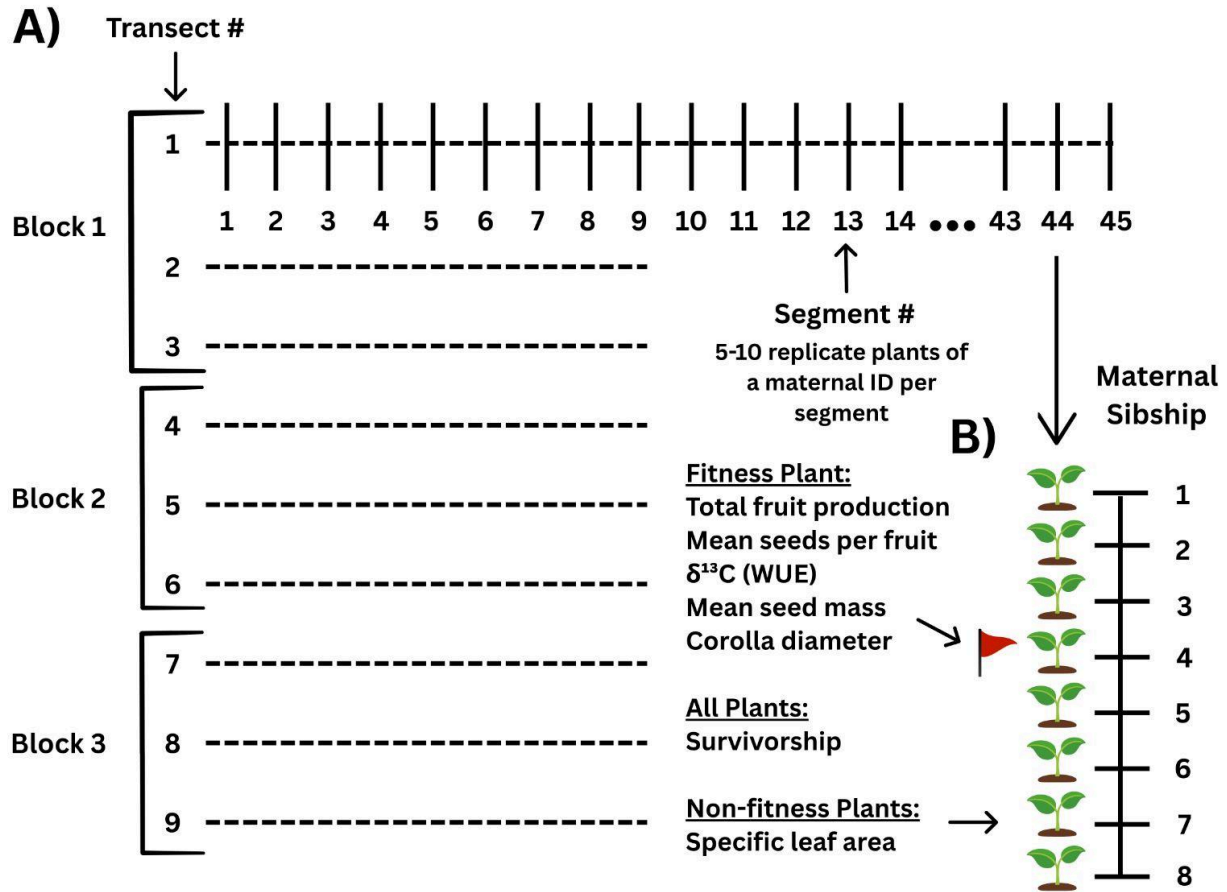


Figure 4. Experimental design at field sites: (A) Each field population was organized into three blocks, and within each block, 3-4 transects were established. Along each transect, segments were positioned at 1-m intervals. (B) In the fall of Year 1, 8-10 seeds were individually sown within each segment, with each maternal family represented once per block (i.e., one segment per block). Each segment contained 8 parental seeds (10 in AC). In Year 2, segments were sown with seeds from two maternal families, with five seeds per family randomly selected from each maternal family. Because block effects were not significant in any site in Year 1, blocks were excluded from subsequent analyses in 2023. To distinguish between the two families sown along the same segment, five seeds from each family were planted on opposite sides of the segment's midpoint. During the flowering periods in winter and spring of Year 1 and Year 2, one individual per segment $\times$ maternal family $\times$ generation was randomly selected as the focal "fitness" plant, from which reproductive trait data were collected when possible.

Statistical Analysis: All analyses were performed in R version 4.2.3 (R Core Team, 2023).

Plasticity was estimated in each of the four focal populations (AC, BB, BO, and HR).

Estimating trait-specific plasticity: Phenotypic plasticity was assessed as the difference in trait expression between Years 1 and 2. We used complementary plasticity metrics to distinguish between (i) the magnitude of environmental responsiveness (PIv) independent of direction and (ii) the magnitude and direction of trait change between years (directional plasticity). To quantify the magnitude of phenotypic plasticity independent of the directional change in phenotype between years, we calculated the phenotypic plasticity index based on maximum and minimum means (PIv) for each paternal family and trait (Valladares et al., 2006, *Eq. 1*):

$$Eq. 1 \text{ PIv} = \frac{(\text{maximum year-specific mean}) - (\text{minimum year-specific mean})}{(\text{maximum year-specific mean})}$$

where the “*maximum*” and “*minimum year-specific means*” refer to the higher and lower of the two mean trait values expressed by a given paternal family between the two study years (2022 and 2023). Thus, PIv captures the proportional change in mean trait expression between years, with values of zero indicating no plasticity and larger values indicating greater plasticity.

Because some traits (SLA and mean seed mass) exhibited mild right-skew, we additionally calculated the corresponding median-based index (PImd) to assess robustness to non-normal trait distributions (Valladares et al., 2006). Plasticity estimates derived from PIv and PImd were strongly correlated in all populations ($r \approx 0.63\text{-}0.94$, Table S4); therefore, PIv was used to estimate the magnitude of plasticity and reflects the extent to which a paternal family altered its phenotype between years, independent of the direction of change.

Directional plasticity was quantified for each trait as the proportional change in the mean phenotypic value from Year 1 to Year 2. This was calculated by taking the difference between the mean phenotypic value of a paternal family in Year 2 and Year 1, divided by the mean phenotype in Year 1 (*Eq. 2*).

$$Eq. 2 \text{ } \Delta(\text{trait phenotype}_{\text{Year 1} \rightarrow \text{Year 2}}) = \frac{(\text{Mean phenotype in Year 2} - \text{Mean phenotype in Year 1})}{\text{Mean phenotype in Year 1}}$$

Positive values indicate an increase in trait values from Year 1 (2022) to Year 2 (2023), whereas negative values indicate a decrease.

Comparing trait-specific plasticity: To determine whether traits differed in their mean magnitude of phenotypic plasticity, the mean PIV value of all paternal families was calculated for each trait. This trait-level mean PIV provides a standardized measure of how plastic each trait was, on average across families, independent of the direction of phenotypic change within or among paternal families.

Additionally, to evaluate whether mean trait expression differed between years, we fit separate linear mixed-effects models for each focal trait within each population (*Eq. 3*).

$$Eq. 3 \text{ trait} \sim \text{Year} + (1|\text{paternal family}) + (1|\text{maternal family})$$

In these models, Year was included as a fixed effect representing differences in environmental conditions between 2022 and 2023, while paternal family and maternal family were included as random intercept effects to account for non-independence among related individuals. Significant fixed effects of Year indicated that average trait values differed between years, providing evidence of environmentally induced shifts in trait expression. The proportion of phenotypic variance explained by Year was further quantified using marginal R^2 values, whereas conditional R^2 values represented variance explained by the full model, including both fixed and random effects.

Estimating multivariate plasticity: To estimate overall multivariate plasticity, we averaged trait-specific PIV values among all focal traits for each paternal family. This metric represents the mean magnitude of plasticity expressed by a paternal family across traits. Larger values indicate greater overall multivariate phenotypic change between years, whereas values near zero indicate relatively stable trait expression across environmental conditions.

Testing genotype-by-year interactions: To evaluate whether paternal families differed in their plastic responses to environmental variation between years, we tested for genotype-by-year ($G \times Y$) interactions using linear mixed-effects models fit with the lmer function in the lme4 package (version 1.1-35, Bates et al., 2015) in R. A significant $G \times Y$ interaction indicates that paternal families differ in their phenotypic responses between years, such that the change in trait expression between years varies among families. Year was included as a categorical fixed effect to estimate population-level change in mean trait values between years. To test for $G \times Y$ interactions, we compared two models for each trait: a null model including random intercepts for paternal and maternal families (Eq. 3), and a full model that also included a paternal family $\times$ Year random effect (Eq. 4).

$$\text{Eq. 4 trait} \sim \text{Year} + (1|\text{paternal family}) + (1|\text{maternal family}) + (1|\text{paternal family:Year})$$

The parental family $\times$ Year interaction term was included as a random effect to estimate variation among paternal families in their plastic response to the environmental difference between years. In other words, it detects whether the magnitude or direction of the phenotypic response to Year differed among paternal families. The significance of the $G \times Y$ term was assessed by comparing models using a chi-squared likelihood ratio test. Variance components were extracted from the full model fit and expressed as both raw variance, the standard deviation of the variance component, and the percentage of total phenotypic variance attributable to the variance component in order to compare the relative contribution of each random effect. Because the null hypothesis that the parental family $\times$ Year variance component equals zero

($\sigma^2_{\text{parental year}} = 0$) lies on the boundary of the parameter space, reported p-values are half of the standard chi-squared p-values (Self and Liang, 1987)

Estimating fitness: To estimate fitness, we used aster models to jointly model sequential life-history stages, each characterized by distinct statistical distributions (Payne et al., *in prep.*, Geyer et al., 2007; Shaw et al., 2008; Geyer, 2023). Our graphical model included survival to flowering (Bernoulli), designation as the fitness plant (Bernoulli), total lifetime fruit production (Poisson), presence of closed fruit (Bernoulli), and mean seeds per fruit (Poisson). The mean fitness ($\overline{W}$) of a paternal family was calculated as the expected number of filled seeds produced by an individual of that paternal family, unconditional on intermediate life history stages (Payne et al., *in prep.*).

Transect was included as a continuous fixed effect at sites where applicable to account for spatial environmental gradients, and location was included as a continuous fixed effect at BB. When modeled independently, both maternal and paternal random effects explained significant and comparable portions of variation in W ; however, when included simultaneously, neither effect remained significant. We therefore combined these components into a single random effect, hereafter referred to as “parental variance,” following Peschel and Shaw (2024), Geyer (2022), and Falconer and Mackay (1996). Model comparisons were conducted using likelihood-ratio tests.

Aster models were fit separately for each year, allowing independent estimation of W and sire effects on W . Sire effects on fitness provide an estimate of additive genetic contributions to offspring fitness, estimated from aster models using the half-sibling breeding design. These sire effects on W were therefore used as our fitness metric in subsequent analyses relating plasticity to fitness and are hereafter referred to as fitness (W). Because W integrates survival and reproduction into a single measure of total seed production, variation in the parental effect on W reflects the differences among parental families in their ability to produce successful offspring.

By relating trait plasticity to W , we tested whether parental families that exhibit stronger (or weaker) plastic responses also achieve higher fitness, thereby linking phenotypic plasticity to its potential evolutionary consequences. Correlations involving paternal effects do not account for within-family uncertainty in these estimates.

Testing associations between fitness and plasticity: We quantified the relationship between W and plasticity in two ways. The first approach included determining whether trait-specific directional plasticity predicted fitness in the second year when controlling for genotypic fitness in the first year. We did this by testing the following model to detect the relationship between plasticity and fitness (in Year 2) while controlling for baseline differences in paternal family performance in Year 1 (*Eq. 5*):

$$\text{Eq. 5} \quad W_{\text{Year 2}} = W_{\text{Year 1}} + \text{Plasticity}_{\text{Year 1} \rightarrow \text{Year 2}}$$

Second, for each trait, we also estimated the bivariate correlation among paternal families between directional plasticity and mean parental $\bar{W}$ (calculated for each paternal family as the mean W between Year 1 and Year 2). Correlations were assessed using Pearson's correlation coefficients, and statistical significance was evaluated using two-tailed tests. Both of these approaches enable us to determine whether plasticity is associated with either year-specific or mean fitness. Because mean fitness was estimated in only two years, and because the estimates weighted the year-specific fitnesses equally, our estimates of mean genotypic fitnesses may not be equal to those expressed over the long-term environmental conditions experienced by each population. Ideally, estimates of mean fitness would be weighted by the relative frequency of different environmental conditions across years. We also conducted Pearson's correlations between W in Year 2 and directional plasticity; these results are reported in the supplementary material (Table S15).

Testing associations between the interannual change in fitness and plasticity: In order to assess whether plasticity in each trait is associated with a directional change in W between years, we used two methods. In both approaches, the change in fitness between years ($\Delta W_{Year\ 1 \rightarrow Year\ 2}$ hereafter referred to as ΔW), was calculated as the difference between W in Years 1 and 2 ($W_{Year\ 2} - W_{Year\ 1}$).

First, to evaluate whether plasticity in each trait was associated with changes in fitness independent of overall paternal family performance, we fit a linear model for each trait in which ΔW was the response variable and $\bar{W}$ between years and directional plasticity were the predictor variables (Eq. 6).

$$Eq. 6 \Delta W \sim \bar{W} + \text{directional plasticity}$$

This model allowed us to assess whether plasticity explained variation in ΔW after accounting for differences in overall paternal family performance.

For our second approach, we used linear regression to evaluate the relationship between changes in mean fitness (ΔW ; response variable) and mean multivariate plasticity (PIv; predictor variable) among paternal families. Models initially included both linear and quadratic terms (Eq. 7):

$$Eq. 7 \Delta W \sim \text{mean_PIv} + I(\text{mean_PIv}^2)$$

where the quadratic term (mean_PIv^2) was included to evaluate potential nonlinear relationships between plasticity and fitness. Significant quadratic relationships would indicate that intermediate levels of plasticity were associated with the greatest fitness change, independent of the direction of plasticity. When the quadratic term was not significant, simplified linear models were used (Eq. 8):

$$Eq. 8 \Delta W \sim \text{mean_PIv}$$

Pearson's correlations between ΔW and both PIV and directional plasticity were also conducted; these results are reported only in the supplementary material (Table S16, Table S17).

Sensitivity Analyses: As a sensitivity analysis, we also estimated W as the mean observed fecundity (total fruits $\times$ mean seeds per fruit) per paternal family and year, which avoids modelling assumptions and is consistent with the fitness metric used in Chapter 2 (Payne et al., in prep)

Results:

Environmental conditions: Environmental conditions differed between years, with substantially lower cumulative precipitation in Year 1 than in Year 2 at all sites, while mean annual temperatures and vapor pressure deficit remained similar between years (Table 2, Figure S5 and S6).

Table 2. Mean annual temperature (T, °C), cumulative precipitation (PPT, mm), and mean vapor pressure deficit (VPD, kPa) during Year 1 and Year 2 ($\pm$ SD), at each study site: Angelo Coast (AC), Bodega Bay (BB), Blue Oak (BO), and Hastings Reserve (HR). Data were obtained from real-time environmental sensors installed at each field station and archived in the historical Dendra records. Year 1 was drier than Year 2 at all sites.

Site	Cumulative annual PPT (mm) Year 1	Mean annual T (°C) Year 1	Cumulative annual PPT (mm) Year 2	Mean annual T (°C) Year 2	Mean VPD (kPa) Year 1	Mean VPD (kPa) Year 2
AC	1,410.0	11.0	1,963.2	11.1	0.29 ($\pm$ 0.39)	0.34 ($\pm$ 0.59)
BB	617.5	11.1	1,147.1	11.5	0.20 ($\pm$ 0.20)	0.14 ($\pm$ 0.18)
BO	421.4	13.8	1,010.4	12.7	0.61 ($\pm$ 0.73)	0.40 ($\pm$ 0.47)
HR	374.4	13.7	953.0	12.7	0.70 ($\pm$ 0.76)	0.48 ($\pm$ 0.57)

Trait-specific plasticity: The magnitude of phenotypic plasticity differed among traits and populations (Table S6). In all four populations, mean seed mass exhibited the highest PIV values

(AC = 0.16 ± 0.02 , BB = 0.269 ± 0.028 , BO = 0.283 ± 0.029 , HR = 0.216 ± 0.018), whereas $\delta^{13}\text{C}$ consistently exhibited low PIV values (AC = 0.02 ± 0.002 , BB = 0.021 ± 0.002 , BO = 0.046 ± 0.005 , HR = 0.012 ± 0.001). SLA and corolla diameter displayed intermediate levels of PIV that differed among populations. Paternal families exhibited substantial heterogeneity in the magnitude of phenotypic plasticity for each trait, with family-level PIV values spanning approximately an order of magnitude (AC: 0.02–0.17, BB: 0.05–0.27, BO: 0.00–0.25, HR: 0.01–0.20; Table S6). Reaction norms revealed substantial heterogeneity among paternal families in both the magnitude and direction of trait responses between years (Figure 5). Similarly, multivariate trait-space trajectories of SLA and $\delta^{13}\text{C}$ indicated that paternal families within populations often shifted in similar directions between years, despite differences in response magnitude among paternal families (Figure 6).

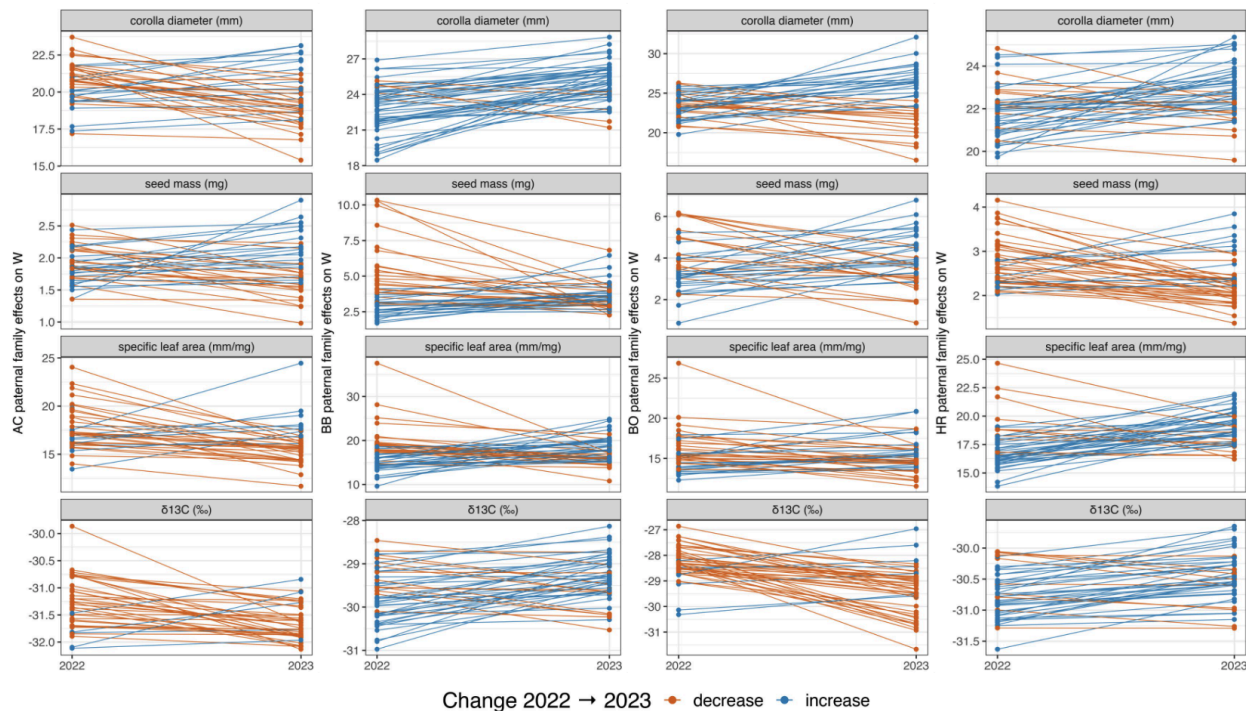


Figure 5. Reaction norms showing trait changes from Year 1 to Year 2 for paternal families; each line represents a paternal family. Reaction norms show paternal family changes in the mean phenotype of four functional traits (corolla diameter, seed mass, specific leaf area (SLA), and

$\delta^{13}\text{C}$) from Year 1 to Year 2 in each population. Each line represents a paternal family measured in both Years, with blue lines indicating trait increases and orange lines indicating decreases.

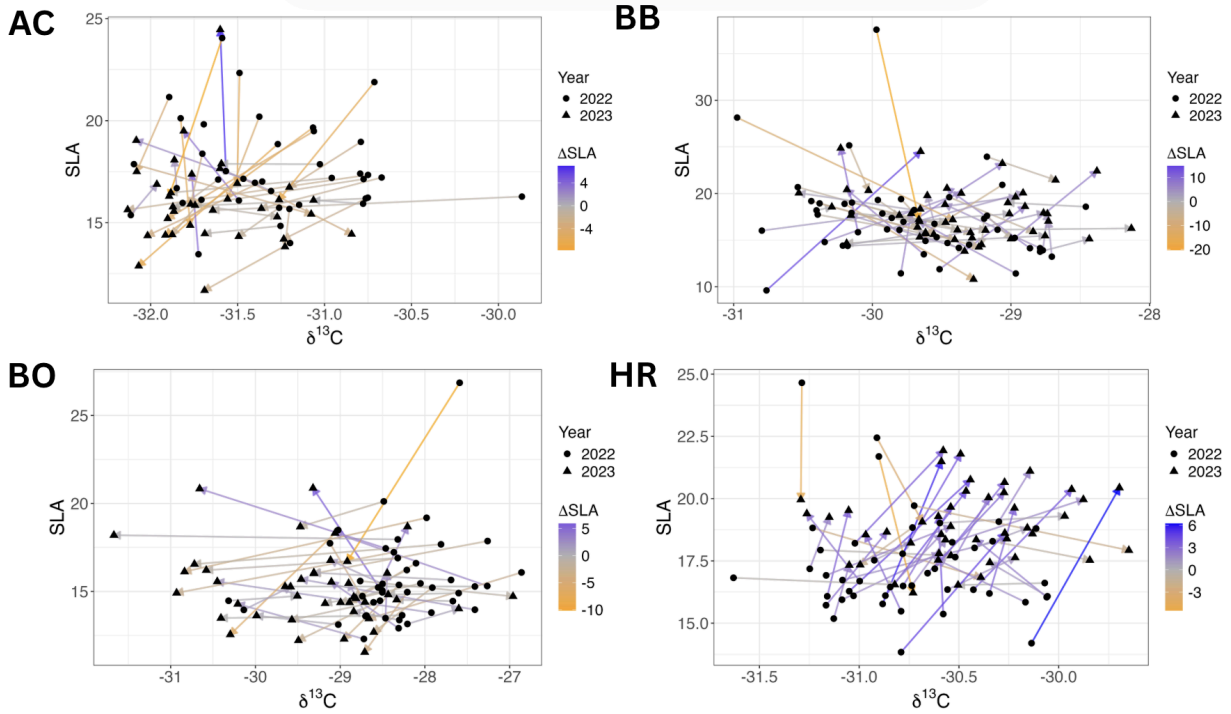


Figure 6. Multivariate plasticity trajectories for SLA and $\delta^{13}\text{C}$ among paternal families in four populations of *N. menziesii*. Points represent paternal family mean trait values in Year 1 (2022; circles) and Year 2 (2023; triangles) for specific leaf area (SLA) and $\delta^{13}\text{C}$ within each population (AC = Angelo Coast, BB = Bodega Bay, BO = Blue Oak, HR = Hastings Reserve). Arrows connect the same paternal family across years and illustrate the direction and magnitude of coordinated trait change between years. Arrow color indicates the magnitude and direction of change in SLA (ΔSLA), with blue indicating increases and orange indicating decreases from 2022 to 2023.

Although all traits exhibited moderate to high PIV values, mixed-effects models revealed that directional changes in population mean trait values between years were not always significant (Table S8). Among traits, $\delta^{13}\text{C}$ consistently exhibited the largest marginal R^2 values in each population (AC: 0.054, BB: 0.016, BO: 0.167, HR: 0.038), indicating that year-specific environmental conditions explained a relatively greater proportion of variation in $\delta^{13}\text{C}$ than in other traits. Corolla diameter and SLA also shifted significantly between years in several populations, although Year generally explained only a modest proportion of total phenotypic

variance. In contrast, mean seed mass showed little evidence of consistent directional change between years despite relatively high PIV values.

Genotype-by-Year Interactions: Likelihood ratio tests revealed no significant genotype-by-year ($G \times Y$) interactions for any trait in any population (all $p > 0.05$). Consistent with this finding, variance components attributed to the paternal family:Year random effect were negligible or zero for the majority of trait-population combinations (Figure 7, Table S9). Where $G \times Y$ variance was non-zero, it remained small in magnitude: the largest contributions were observed for $\delta^{13}C$ in BO (8.81%) and AC (7.99%), and for mean seed mass in HR (7.08%) and BB (4.12%). $G \times Y$ variance was absent entirely for SLA in all four populations and for corolla diameter in all populations except AC, where it accounted for 5.19% of total variance. For all traits and in all populations, residual variance was the dominant source of phenotypic variation, consistently explaining more than 70% of total variance and exceeding 90% in many cases (e.g., SLA: 95–100%; corolla diameter: 83–97%). The absence of significant $G \times Y$ interactions indicates that paternal families responded similarly to environmental variation.

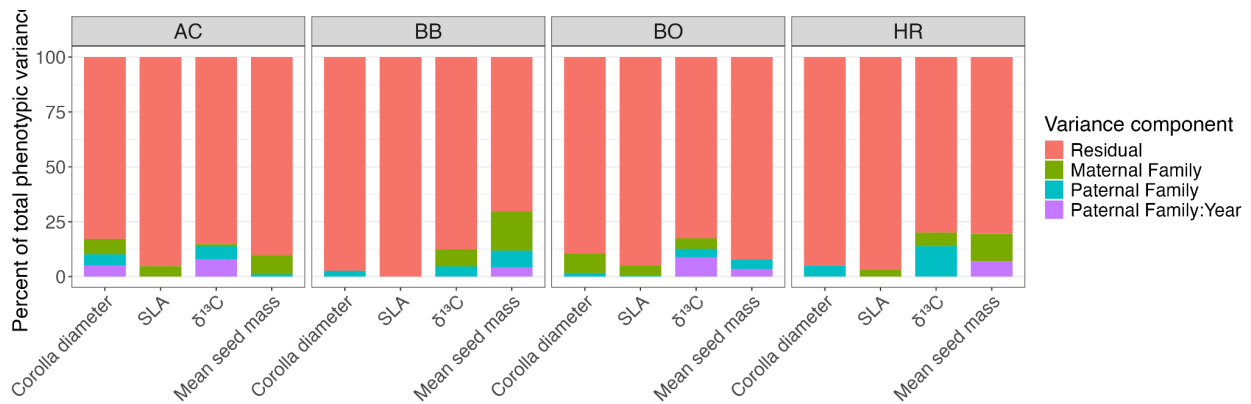


Figure 7. Variance partitioning of trait expression in each population. Stacked barplots show the proportion of total phenotypic variance attributable to paternal family, maternal family, paternal family:Year ($G \times Y$), and residual variance for each trait (corolla diameter, specific leaf area (SLA), $\delta^{13}C$, and mean seed mass) in each population (AC: Angelo Coast, BB: Bodega Bay, BO: Blue Oak, HR: Hastings Reserve).

Associations Between Fitness and Plasticity: Effects of directional plasticity on fitness were generally weak and inconsistent among traits and populations (Table S11; Table S12). After accounting for W in Year 1, linear models indicated an effect of directional plasticity on W in Year 2 in only two cases: At BO, increases in corolla diameter between years 1 and 2 were associated with higher W in Year 2 ($\beta = 0.24$, $p = 0.019$), and in HR, increases in mean seed mass were positively associated with W in Year 2 ($\beta = 0.90$, $p = 0.045$). Multivariate plasticity exhibited a weak positive association with W in BO ($\beta = 0.16$, $p = 0.067$), while directional plasticity in $\delta^{13}\text{C}$ in HR also showed a weak positive relationship with W ($\beta = 17.0$, $p = 0.061$). No other trait-specific or multivariate plasticity metrics were significantly associated with W in Year 2.

Pearson's correlations between estimated $\overline{W}$ and directional plasticity revealed similarly weak relationships in each population (Figure 8, Table S12). Most correlations were small and non-significant, although multivariate plasticity in AC exhibited a marginally nonsignificant negative relationship with $\overline{W}$ ($r = -0.31$, $p = 0.052$). Moreover, in HR, directional plasticity in mean seed mass was positively correlated with parental $\overline{W}$ ($r = 0.29$, $p = 0.049$), while directional plasticity in $\delta^{13}\text{C}$ showed a weak positive correlation ($r = 0.27$, $p = 0.077$).

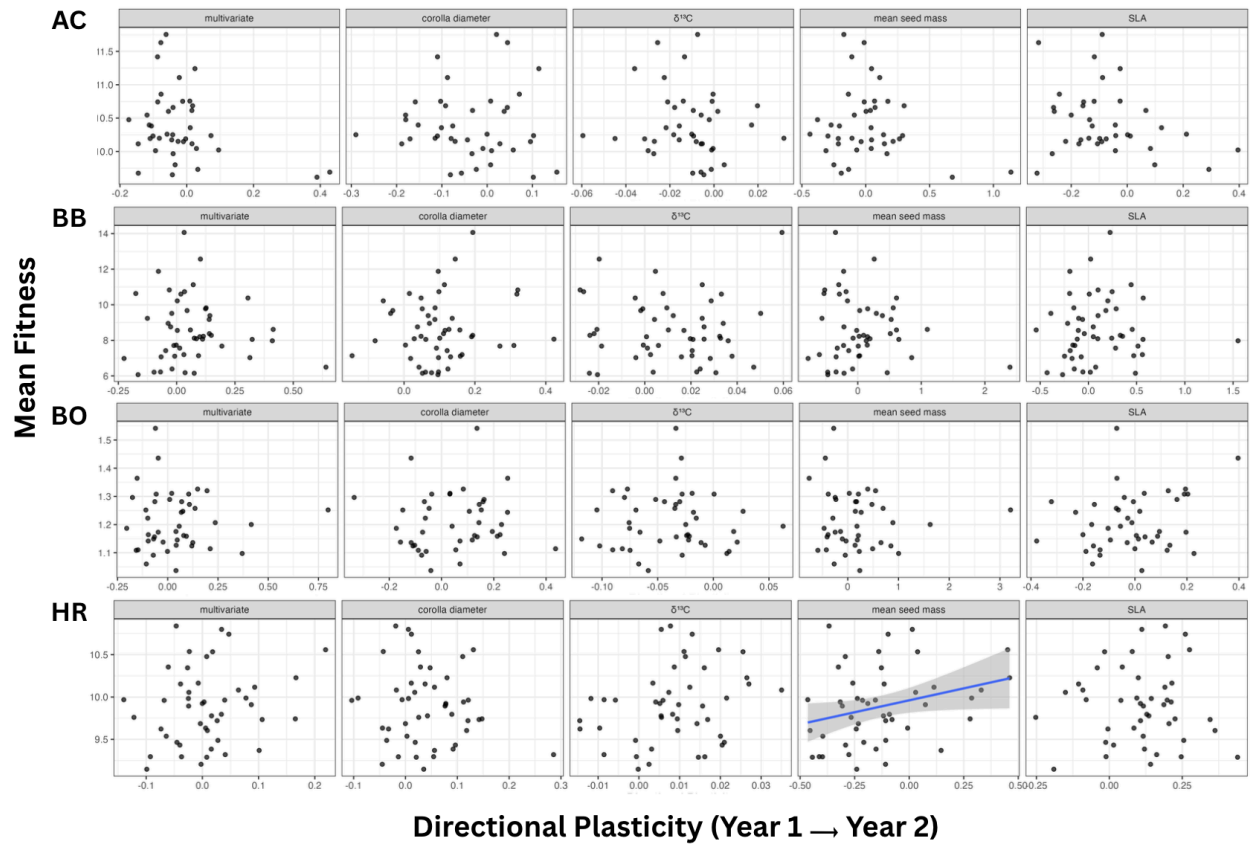


Figure 8. Relationships between directional plasticity and mean fitness ($\bar{W}$) (i.e., the paternal effect on fitness) among paternal families in each of four populations of *N. menziesii*.

Scatterplots show associations between directional plasticity and estimated $\bar{W}$ for five plasticity metrics: multivariate plasticity, and the plasticity of corolla diameter, $\delta^{13}\text{C}$, mean seed mass, and specific leaf area (SLA). Directional plasticity was calculated as the proportional change in mean trait values between Year 1 and Year 2 for each paternal family. Each point represents a paternal family. Populations are shown by row (AC = Angelo Coast, BB = Bodega Bay, BO = Blue Oak, HR = Hastings Reserve). Blue regression lines indicate statistically significant relationships ($p < 0.05$); no regression lines are shown for non-significant relationships.

Associations Between Fitness Change and Plasticity: Associations between the absolute magnitude of plasticity (PIv) and ΔW were largely absent (Table S13; Table S18). Most relationships between PIv and ΔW were best fit by linear models and were not significantly different from zero. However, in HR, greater plasticity in corolla diameter was negatively associated with ΔW ($\beta = -6.0$, $p = 0.014$); paternal families exhibiting relatively large changes in corolla diameter experienced greater declines in mean fitness between years than families with

relatively stable corolla diameters. At BB, quadratic relationships were detected between ΔW and plasticity in both $\delta^{13}C$ ($\beta = 4078.36$, $p = 0.013$) and mean seed mass ($\beta = -21.65$, $p = 0.05$), suggesting nonlinear associations between plasticity and ΔW . No other trait-specific or multivariate plasticity metrics were significantly associated with ΔW .

Regression models evaluating directional plasticity while accounting for $\overline{W}$ revealed similarly limited evidence for consistent associations with ΔW (Table S14). In BB, greater positive directional plasticity in $\delta^{13}C$ was positively associated with ΔW between years ($\beta = 18.0$, $p = 0.001$), indicating that paternal families exhibiting relatively large increases in $\delta^{13}C$ also tended to exhibit greater increases in $\overline{W}$ between years. In BO, greater positive directional plasticity in corolla diameter ($\beta = 0.56$, $p = 0.010$), mean seed mass ($\beta = 0.10$, $p = 0.043$), and multivariate plasticity ($\beta = 0.39$, $p = 0.034$) were positively associated with ΔW , indicating that paternal families exhibiting relatively large increases in these traits between years tended to experience smaller declines in mean fitness. No significant relationships were detected for AC or HR. Overall, these results provide insufficient evidence that either the magnitude or direction of plasticity was consistently associated with ΔW .

Sensitivity Check: Sensitivity analyses using family-mean fecundity in place of aster sire effects yielded similarly weak and inconsistent associations between plasticity and fitness. Notably, the specific trait-population combinations that reached $p < 0.05$ differed between the two fitness metrics (Table S20), indicating that the isolated significant associations observed in either analysis are unlikely to reflect consistent biological effects and may instead represent false positives under the multiple-testing burden of 4 traits $\times$ 4 populations $\times$ multiple model frameworks.

Discussion:

Phenotypic plasticity is a key mechanism by which plants respond to environmental variation, yet its adaptive significance depends on how plastic responses influence W in different environments. In this study, we found that plasticity in *N. menziesii* differed among traits and between populations, indicating that environmentally induced phenotypic variation is context-dependent, although some traits show more consistent plastic responses ($\delta^{13}\text{C}$) among populations than others (mean seed mass). We found little evidence for $G \times Y$ interactions, suggesting that our four study populations harbor low genetic variation in plasticity for our focal traits in the two years examined. Among genotypes, trait-specific plasticity was not consistently associated with W or ΔW . Eriksson et al. (2023) clarified that reaction norms or shifts in trait values across environments are insufficient to infer adaptive plasticity without explicitly linking those changes to W outcomes. Nevertheless, plasticity remains an important strategy by which plants can respond to environmental change, potentially contributing to short-term tolerance under changing conditions (Pigliucci, 2001; Loreto and Atzori, 2024).

Different traits respond uniquely to environmental conditions: Based on reaction norms, each trait responded uniquely to inter-annual variation in local conditions, indicating they may have different developmental and physiological responses to environmental variation. Surprisingly, less negative values of $\delta^{13}\text{C}$, which are commonly interpreted as indicating higher WUE (Bchir et al., 2016) occurred at two of the sites (BB and HR) in the wetter year (Year 2). By contrast, in these populations, the more mesic conditions of Year 2 induced the production of leaves with higher SLA, corresponding to thinner, more acquisitive leaves, which are often considered advantageous (compared to leaves with low SLA) under relatively mesic conditions. Consequently, environmentally induced changes in WUE at BB and HR opposed the adaptive prediction that mesic conditions would favor lower WUE, whereas SLA exhibited plastic

responses in the predicted direction. The opposite trends were observed at AC and BO for both SLA and WUE.

These inconsistent observations between populations and traits differ from previous reports that lower SLA (thicker leaves) is associated with less negative values of $\delta^{13}\text{C}$ and higher WUE both within and among species (Rao and Wright, 1994; Lamont et al., 2002; Hoffmann et al., 2005), indicating a more tightly integrated leaf economic strategy. The lack of such integration reported here indicates that traits commonly assumed to be correlated along a conservative-acquisitive spectrum may respond independently to environmental drivers in natural populations. Additionally, we observed increases in corolla diameter at BB and HR in Year 2 alongside less negative $\delta^{13}\text{C}$ values, suggesting that paternal families responded to more mesic conditions by producing larger corolla while also exhibiting higher WUE. This pattern contrasts with findings in *Mimulus*, where larger flowers were associated with lower WUE (Kelly et al., 2008), with the expectations that larger flowers may increase evaporative water loss due to the high transpiration demand of floral tissues (Galen et al., 1999; Lambrecht and Dawson, 2007). One possible explanation for trait-specific responses to drought is that focal traits differ in their developmental timing, physiological function, and environmental sensitivities. For example, SLA or $\delta^{13}\text{C}$ reflect leaf structure and water-use physiology during vegetative growth, whereas corolla diameter and seed mass reflect reproductive allocation later in development; these traits may therefore respond to different aspects of water availability or to drought during different life stages. Other recent studies report that functional traits often exhibit decoupled or trait-specific responses to environmental variation rather than coordinated shifts along a single conservative acquisitive axis (Zhang et al., 2020; Anderson et al., 2021; Sommer et al., 2025).

Genotype-by-year interactions were generally not a significant source of trait variation: We found no significant G×Y interactions for any trait in any population; G×Y variance components were negligible or absent in the majority of trait-population combinations. Both paternal family and paternal family-by-year variance components were generally small, indicating limited genetic variation among paternal families in average trait values and trait plasticity between years.

The predominance of residual variance among all traits suggests that fine-scale, within-group variation represents the largest source of phenotypic variation in this system. This could potentially be attributable to microenvironmental heterogeneity, developmental stochasticity, or unmeasured sources of dominance or epigenetic variation (Lloyd, 2008; Vogt, 2015; Biwer et al., 2020). Where non-zero G×Y variance components were detected (for $\delta^{13}\text{C}$ in AC and BO and mean seed mass at HR and BB), they were not statistically significant. These results suggest that much of the phenotypic variation in this system may reflect environmental and stochastic sources of variation, with weak evidence for genetically based differences in plastic responses among paternal families.

Traits differ in their magnitude of phenotypic plasticity: Trait-specific differences in PIV suggest that some functional traits are more environmentally sensitive than others, potentially reflecting differences in the developmental and physiological constraints underlying these traits. Despite moderate to high PIV values for several traits, mixed-effects models indicated that year-specific environmental conditions explained only a modest proportion of total phenotypic variance for most traits, with $\delta^{13}\text{C}$ exhibiting the strongest and most consistent year effects in each population. Similar patterns of trait-specific plasticity have been documented in other plant

species, in which traits vary in their sensitivity to environmental change and in their capacity for plastic responses (Valladares et al., 2006; Anderson et al., 2021; Vilellas et al., 2021). For example, Vilellas et al. (2021) found that plasticity masked genetic differentiation for vegetative traits but not reproductive traits, indicating that distinct classes of traits may differ in both their environmental sensitivity and evolutionary stability. These results are consistent with prior work that distinguishes plasticity and trait stability, in which some traits are evolutionarily canalized to maintain the same phenotype across different environments while others respond to environmental changes (Alseekh et al., 2025).

Plastic responses are similar among paternal families: Although we detected clear trait-specific plastic responses at the population level, we found weak support for variation among paternal families in multivariate plasticity, suggesting that the mean magnitude of plasticity was similar among them. Consistent with this pattern, $G \times Y$ variance components were generally non-significant, indicating limited genetic variation in plasticity for most traits. However, the presence of significant genotype-by-year ($G \times Y$) interactions in $\delta^{13}\text{C}$ at AC and BO indicates that paternal families can diverge in how they respond physiologically to environmental variation under certain conditions. This pattern was also evident in multivariate trait-space trajectories (Figure 6), where paternal families within each population generally shifted in similar directions between years despite variation in the magnitude of their responses. Overall, these results indicate that environmental variation affected population mean trait expression for our focal traits, but did not reveal consistent differences among paternal families in the magnitude or direction of their plasticity.

The detection of $G \times Y$ in a trait reflects genetic variation in environmental responsiveness and, therefore, the potential for plasticity to evolve under selection (Napier et al.,

2022). The generally low genetic variation in plasticity observed here suggests that the capacity for plasticity to evolve may be limited for many of the focal traits, at least under the environmental conditions observed. Although estimates of plasticity were based on differences in trait expression between only two years, our robust nested pedigree design strengthens the conclusion that paternal families responded similarly to interannual environmental differences in $\delta^{13}\text{C}$, mean seed mass, corolla diameter, and specific leaf area in all four populations (Nussey et al., 2007). $G \times Y$ interactions can constrain directional selection by altering genotype rankings across environments, potentially contributing to the maintenance of additive genetic variance through fluctuating selection (Sultan and Bazzaz, 1993; Shaw et al., 1995). The lack of strong $G \times Y$ interactions indicate that interannual environmental differences experienced by the field populations in this study did not substantially alter the relative expression of traits among paternal families, implying relatively consistent multivariate responses across years. In sum, limited additive genetic variance in plasticity implies that the capacity for plastic responses to evolve may be restricted, even in the absence of strong $G \times Y$ -mediated constraints on selection (Nussey et al., 2007). Overall, the predominance of parallel multivariate trajectories, weak $G \times Y$ interactions, suggest that environmental variation altered mean trait expression more strongly than relative differences among paternal families, resulting in a shared plastic response among genotypes within a population.

Weak Associations Between Plasticity and Fitness: Plastic responses may be adaptive, maladaptive, or neutral depending on the environmental context and the traits involved (Van Kleunen and Fischer, 2005; Ghalambor et al., 2007). In contrast to studies in which plasticity has been found to be associated with higher fitness or adaptive differentiation among environments (Anderson et al., 2021; Walter et al., 2023), our results indicate that plasticity in *N. menziesii* was

weakly associated with fitness and showed limited genetic variation. Most relationships between directional plasticity and W were small, inconsistent, and statistically nonsignificant, suggesting that environmentally induced changes in trait expression did not reliably increase W under the environmental conditions examined here. Although a small number of population- and trait-specific associations were detected, these relationships were not consistently supported among W metrics and may reflect the large number of statistical comparisons conducted across traits, populations, and model types rather than strong biological effects. In contrast, Zhang et al. (2020), who found that responses to precipitation variation were strongly associated with trait plasticity in alpine grassland plants, plasticity in our study was only weakly and inconsistently associated with fitness.

Despite these isolated associations in *N. menziesii*, trait-specific and multivariate plasticity were not consistently associated with either W or ΔW in any of the four populations examined here. Although some magnitudes or directions of plasticity were associated with higher fitness under particular environmental conditions (Year 1 or Year 2), most trait-specific and multivariate plasticity metrics were unrelated to either W or ΔW , and several significant relationships differed in direction among populations. Quadratic relationships between plasticity and ΔW in BB further suggest that intermediate magnitudes of plasticity were associated with the highest fitness responses, consistent with nonlinear or stabilizing selection between plasticity and fitness. However, none of the patterns observed were consistent among traits or between populations. These results suggest that while some environmentally induced trait changes may be associated with higher fitness under specific environmental conditions, plasticity in the focal traits examined here was generally weakly associated with fitness.

This context dependency is consistent with patterns reported by Sultan (2015), who

emphasizes that phenotypic plasticity can contribute to resilience under environmental change, but that the fitness consequences of plastic responses depend strongly on the ecological conditions experienced by individuals and populations. Rather than universally buffering populations against environmental variation, Sultan (2015) points out that plastic responses may enhance performance in some environments while being neutral or even maladaptive in others, particularly when environmental conditions vary spatially or temporally. Similarly, a review of plant studies by Matesanz et al. (2010) demonstrated that the adaptive value of plasticity frequently depends on environmental context, with plastic responses often differing in both magnitude and fitness consequences among environments and populations. The lack of agreement among fitness metrics on which specific associations were significant further suggests that plasticity had limited and inconsistent adaptive consequences in these populations under year 1 and year 2 environmental conditions.

Plastic responses affecting W could occur in traits not quantified in this study, such as belowground root allocation, phenology, or physiological processes, potentially decoupling the observed trait plasticity in our focal traits from reproductive output (Sultan, 2000; Nussey et al., 2007; Nicotra et al., 2010). Additionally, plastic responses may reflect physiological constraints or trade-offs rather than adaptive adjustments, resulting in trait changes that do not increase (and may even reduce) W under certain conditions. Our results suggest that plastic responses in our focal traits were not consistently associated with fitness among populations and may represent neutral, context-dependent, or occasionally mismatched responses to environmental conditions (Nicotra et al., 2010)

Prior research on genetic correlations among traits: In the previous chapter, we found that genetic correlations among the focal traits examined here were generally weak or undetectable,

indicating a low degree of genetic integration among traits (Payne et al. *in prep*). We also detected moderate to high additive genetic variance for each of these traits, indicating substantial potential for independent evolutionary responses (Payne et al. *in prep*). These findings suggest that genetic correlations among the focal traits within this study may impose relatively limited constraints on short-term evolutionary responses in the four populations examined here, although uncertainty surrounding covariance estimates should be considered when interpreting the magnitude of these constraints. When genetic correlations are strong, they can influence adaptation by causing some traits to evolve indirectly in nonadaptive or maladaptive ways due to their correlated responses to selection on other traits (Lande, 1979; Etterson and Shaw, 2001). The weak or undetectable correlations observed in the four populations studied here indicate that our focal traits may be able to respond to selection independently of each other. Moreover, the weak associations between plasticity and fitness, together with the limited genetic variation detected for plasticity, suggest that environmentally induced trait changes were either weakly associated with fitness differences or may reflect past selection that reduced variation in plastic responses among paternal families.

Conclusion:

By integrating trait plasticity, fitness estimates, and $G \times Y$ analyses in four wild populations of *N. menziesii*, our study demonstrates that environmentally induced phenotypic changes can differ substantially among traits and populations without consistently translating into increased fitness. Notably, $\delta^{13}\text{C}$ exhibited strong and consistent plasticity among populations, suggesting that physiological traits related to water-use-efficiency may be particularly sensitive to environmental variation even when those responses are only weakly associated with fitness outcomes. Overall, our results suggest that phenotypic plasticity in natural

populations may frequently be neutral or context-dependent with respect to fitness rather than consistently adaptive in different environments. In sum, the absence of strong relationships between plasticity and fitness remains a factor that allows this variation to accumulate.

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